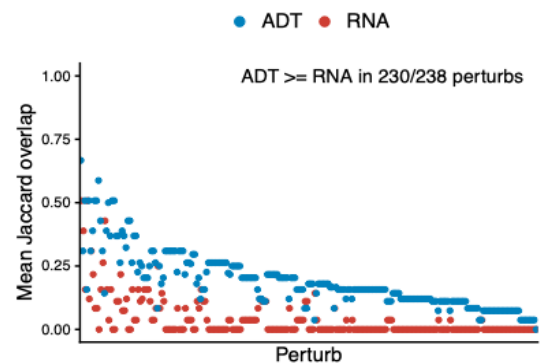
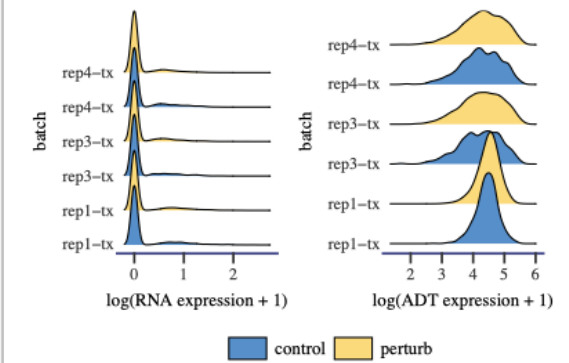
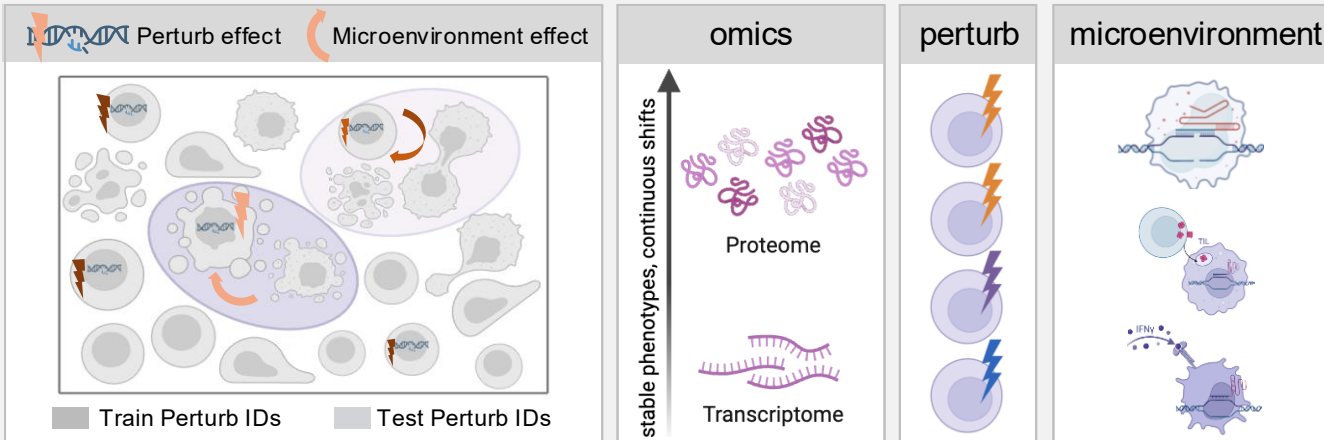


a Motivation: Why multi-omics perturb?

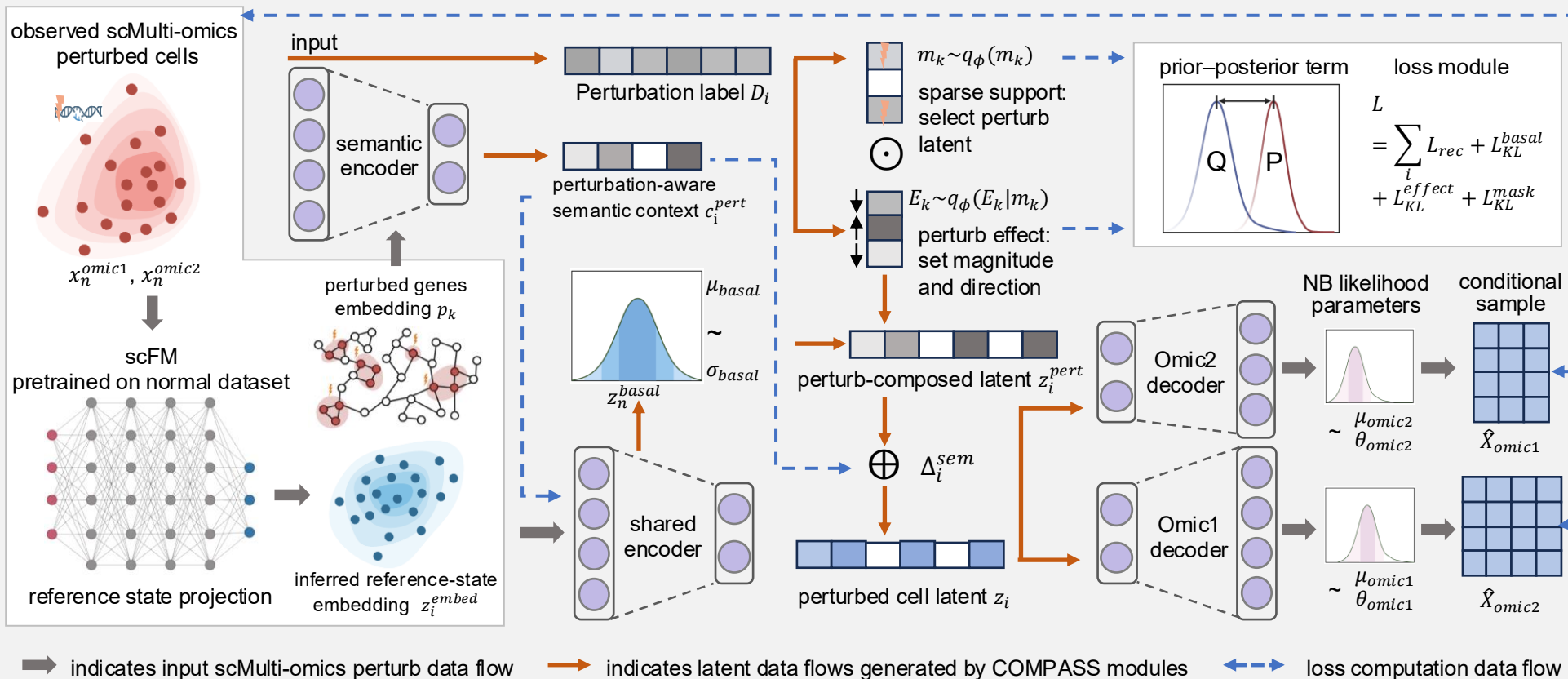
Cross-batch stability of perturb-response



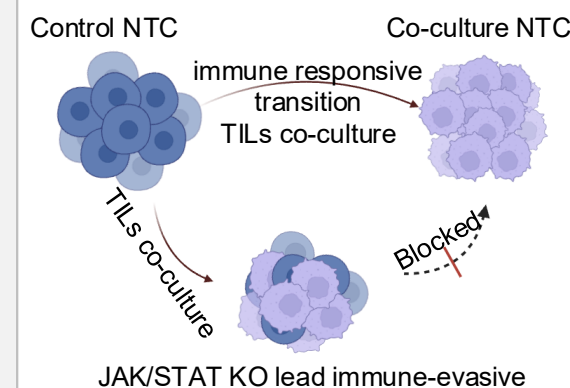
Matched marker consistency across omics

**b** Input: Single cell multi-omics perturbation data**c**

COMPASS algorithm

**d** Downstream analysis

Predicting unseen gene-microenvironment effects to identify immune-state regulators



Recovering regulatory pathway from noisy perturb data

